

Practical Session 10: Portfolios, Views, and Backtests

Hands-On — Efficient Frontier, BL Views, and a Cost-Aware Walk-Forward

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Session Overview

Duration: 1 hour (50 min problems + 10 min debrief)

Agenda:

- ➊ **Recap** (5 min): mean-variance, BL posterior, OOS Sharpe, turnover cost.
- ➋ **Problem 1** (12 min): two-asset frontier — minimum-variance & tangency weights.
- ➌ **Problem 2** (12 min): map an LLM sentiment score to a Black–Litterman view.
- ➍ **Problem 3** (10 min): does a daily signal survive transaction costs?
- ➎ **Case study** (15 min): walk-forward backtest of a sentiment factor in Python (`code/notebooks/10-portfolio-quant-trading/`).
- ➏ **Discussion** (6 min): when is an LLM signal worth deploying?

Goal

Leave able to translate a textual signal into a portfolio weight *and* to judge, quantitatively, whether the resulting strategy is deployable.

Recap: The Four Formulas You Will Use

UNDER THE HOOD

Mean-variance: $\mu_p = \boldsymbol{\mu}^\top \mathbf{w}$, $\sigma_p^2 = \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}$; tangency $\mathbf{w}^{\text{SR}} \propto \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}$.

Black–Litterman posterior mean:

$$\hat{\boldsymbol{\mu}} = [(\tau \boldsymbol{\Sigma})^{-1} + \mathbf{P}^\top \boldsymbol{\Omega}^{-1} \mathbf{P}]^{-1} [(\tau \boldsymbol{\Sigma})^{-1} \boldsymbol{\Pi} + \mathbf{P}^\top \boldsymbol{\Omega}^{-1} \mathbf{q}], \quad q_i = \alpha s_i + \Pi_i.$$

Out-of-sample Sharpe (annualised):

$$\text{SR}^{\text{OOS}} = \frac{\bar{r}_p^{\text{OOS}}}{\hat{\sigma}_p^{\text{OOS}}} \sqrt{252}.$$

Turnover transaction cost:

$$\text{TC}_t = c \sum_i |\Delta w_{i,t}|, \quad \Delta w_{i,t} = w_{i,t} - w_{i,t-1}.$$

Problem 1: Two-Asset Frontier — Setup

Two assets (monthly excess returns), annualise where noted:

$$\boldsymbol{\mu} = \begin{pmatrix} 0.8\% \\ 0.4\% \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} 0.0036 & 0.0009 \\ 0.0009 & 0.0016 \end{pmatrix}.$$

So $\sigma_1 = 6\%$, $\sigma_2 = 4\%$, correlation $\rho = \frac{0.0009}{0.06 \cdot 0.04} = 0.375$.

Tasks:

- A.** Minimum-variance weights for a long-only fully-invested portfolio, $\mathbf{w} = (w, 1 - w)$. Use $w^* = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$.
- B.** The (unconstrained) tangency direction $\mathbf{w}^{\text{SR}} \propto \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$, then normalise to sum 1.
- C.** Compare to $1/N = (0.5, 0.5)$. Which is closest to tangency? Comment in light of DeMiguel et al. (2009).

Problem 2: From LLM Score to a BL View

Setup. An LLM scores firm i 's earnings call: $s_i = +0.6$ (bullish). Equilibrium prior $\Pi_i = 0.30\%$ monthly. Calibration constant $\alpha = 0.8\%$; view-noise scale $\sigma_q^2 = (0.5\%)^2$.

Tasks:

- A.** Compute the view value $q_i = \alpha s_i + \Pi_i$.
- B.** Compute the view uncertainty $\Omega_{ii} = \sigma_q^2(1 - |s_i|)$.
- C.** A second firm scores $s_j = +0.05$ (near-neutral). Compute q_j , Ω_{jj} . Which view will move the posterior more, and why?
- D.** *Conceptual:* you accidentally calibrate α on the same window you evaluate on. Name the bias and the fix.

Hint

Posterior pull toward a view scales with view *precision* Ω_{ii}^{-1} . Strong signal $\Rightarrow$ small $\Omega_{ii} \Rightarrow$ large precision.

Problem 3: Does the Signal Survive Costs?

Setup. A daily long-short sentiment strategy with:

- Gross (pre-cost) annualised return 4.0%, volatility 8.0%.
- Average *one-way* daily turnover 20%; cost $c = 0.04\%$ per unit turnover.
- 252 trading days.

Tasks:

- A.** Annualised transaction cost $\approx \text{turnover} \times 252 \times c$.
- B.** Net annualised return = gross – cost. Net Sharpe (assume vol unchanged).
- C.** By how much must one-way turnover fall to keep net return $\geq 2\%$, holding c fixed?
- D.** *Capacity*: the strategy wants \$10M in a stock with \$50M ADV. What participation rate is that, and does it breach the 5–10% guideline?

Case Study: Walk-Forward Sentiment Backtest

Goal. Run a leakage-free walk-forward backtest of a language factor and read off the out-of-sample, cost-adjusted Sharpe.

Notebook:

`code/notebooks/10-portfolio-quant-trading/demo.ipynb`

Frontier figure: `gen_efficient_frontier.py` (deterministic, no network).

What you will do:

- 1 Build quintile language-factor returns from per-firm sentiment scores.
- 2 Slide the walk-forward window; *fit calibration on train, evaluate on test*.
- 3 Subtract turnover costs; compute SR^{OOS} .
- 4 Answer three diagnostic questions about leakage and decay.

Discipline

Any parameter (α , λ , $W_{\max}$) touched by the test window invalidates the result.

Case Study: The Code

```
import numpy as np, pandas as pd
def language_factor(scores, fwd_ret):          # firms x dates, scores
    in [-1,1]
    """Top-minus-bottom quintile return from demeaned sentiment."""
    s = scores.sub(scores.mean(axis=0), axis=1)    # cross-sectional
        demean
    lf = {}
    for t in s.columns:
        q = s[t].rank(pct=True)
        top, bot = q >= 0.8, q <= 0.2
        lf[t] = fwd_ret.loc[top, t].mean() - fwd_ret.loc[bot, t].
            mean()
    return pd.Series(lf)

def walk_forward_sharpe(lf, train=252, step=21, turnover=0.15, c=5e
-4):
    oos = []
    for m in range(0, len(lf) - train - step, step):
        test = lf.iloc[m + train : m + train + step]    # never seen
            in fit
        oos.append(test - turnover * c)                  # cost-adjust
    r = pd.concat(oos)
    return float(r.mean() / r.std() * np.sqrt(252))
# sharpe = walk_forward_sharpe(language_factor(scores, fwd_ret))
```

Case Study: Diagnostic Questions

Run the notebook, then answer:

Q1 (leakage). The demean step uses `scores.mean(axis=0)` — a *cross-sectional* (per-date) mean. Why is that safe, whereas a *time-series* mean over all dates would leak?

Q2 (decay). Replace raw scores with EWMA-decayed scores ($\lambda \in \{0.1, 0.3, 0.5\}$). How does SR^{OOS} move? Relate to the 1–7 day half-life rule.

Q3 (costs). Sweep $c \in \{0, 5, 10\}$ bps. At what cost does the Sharpe cross below the 1.5 paper-trading threshold? Connect to the capacity constraint.

Optional extension

Feed the per-firm view $q_i = \alpha s_i + \Pi_i$ into a Black–Litterman optimizer instead of the raw quintile sort. Does BL's prior shrinkage raise or lower turnover? Why?

Discussion: Is This Signal Worth Deploying?

For each scenario, decide *go* / *no-go* and justify with the lecture's criteria.

- 1 A large-cap US equity sentiment signal: OOS Sharpe 1.7 in-backtest, but decays within 2 days and the desk runs \$2B. Crowding? Capacity?
- 2 A small-cap emerging-market filing signal: OOS Sharpe 1.2, low coverage, wide spreads (tens of bps). Grossman–Stiglitz niche?
- 3 A real-time 8-K risk-alert system: no return claim, but cuts tail losses. How would you value it, and audit it under SR 11-7?

Frame your answer around

OOS Sharpe *and* T^* , transaction costs, capacity / ADV, crowding, auditability, and whether the LLM is acting as an extractor or (wrongly) as an optimizer.

Solution: Problem 1 — Two-Asset Frontier

A. Minimum-variance ($\sigma_1^2 = 0.0036$, $\sigma_2^2 = 0.0016$, $\sigma_{12} = 0.0009$):

$$w^* = \frac{0.0016 - 0.0009}{0.0036 + 0.0016 - 2(0.0009)} = \frac{0.0007}{0.0034} \approx 0.206 \Rightarrow \mathbf{w}_{\min} \approx (0.21, 0.79)$$

B. Tangency direction $\Sigma^{-1}\boldsymbol{\mu}$ ($\det = 4.95 \times 10^{-6}$):

$$\begin{pmatrix} 0.0016 & -0.0009 \\ -0.0009 & 0.0036 \end{pmatrix} \begin{pmatrix} 0.008 \\ 0.004 \end{pmatrix} = \begin{pmatrix} 9.2 \\ 6.7 \end{pmatrix} \times 10^{-6} \Rightarrow \mathbf{w}^{\text{SR}} \approx (0.58, 0.42).$$

C. $1/N = (0.5, 0.5)$ sits very close to tangency $(0.58, 0.42)$ — and far cheaper to estimate. With noisy $\boldsymbol{\mu}$, the small optimization gain is easily swamped by estimation error (DeMiguel et al., 2009).

Solution: Problem 2 — LLM Score to BL View

Firm i ($s_i = 0.6$):

$$q_i = 0.8\% \times 0.6 + 0.30\% = 0.48\% + 0.30\% = \mathbf{0.78\%}, \quad \Omega_{ii} = (0.5\%)^2(1 - 0.6) =$$

Firm j ($s_j = 0.05$):

$$q_j = 0.8\% \times 0.05 + 0.30\% = \mathbf{0.34\%}, \quad \Omega_{jj} = (0.5\%)^2(1 - 0.05) = \mathbf{2.375 \times 10^{-4}}$$

C. View i moves the posterior more: it has both a larger deviation from the prior (0.78% vs. 0.34%) *and* higher precision (Ω_{ii}^{-1} is $\approx 2.4 \times$ larger). The near-neutral signal j lets the prior dominate — exactly the intended behaviour of $\Omega_{ii} = \sigma_q^2(1 - |s_i|)$.

D. Calibrating α on the evaluation window is *model-selection look-ahead leakage*. Fix: estimate α only on data predating each test window (walk-forward).

Solution: Problem 3 — Survive Costs?

A. Annualised cost:

$$0.20 \times 252 \times 0.04\% = 0.20 \times 252 \times 0.0004 = \mathbf{2.016\% \text{ p.a.}}$$

B. Net return / Sharpe:

$$\text{net} = 4.0\% - 2.016\% \approx \mathbf{1.98\%}, \quad \text{net SR} = \frac{1.98\%}{8.0\%} \approx \mathbf{0.25} \text{ (vs. gross 0.50)}.$$

Costs *halve* the Sharpe — a marginal signal is nearly destroyed.

C. Turnover for net $\geq 2\%$: need cost $\leq 2.0\%$, i.e.

$$\text{turnover} \times 252 \times 0.0004 \leq 0.02 \Rightarrow \text{turnover} \leq \frac{0.02}{0.1008} \approx \mathbf{19.8\%}.$$

Barely below the current 20% — essentially no slack; the strategy is cost-bound.

D. Capacity: \$10M / \$50M ADV = **20%** participation — *double* the 5–10% guideline. Expect material price impact; scale the book down.

Wrap-Up: Key Takeaways

- **P1:** On two assets, $1/N$ already sits near tangency — the optimization prize is small once μ is estimated.
- **P2:** The map $q_i = \alpha s_i + \Pi_i$, $\Omega_{ii} = \sigma_q^2(1 - |s_i|)$ turns a raw LLM score into a confidence-weighted BL view; neutral signals defer to the prior.
- **P3:** Transaction costs and turnover are first-order — a 4% gross strategy can net 2% and lose half its Sharpe; watch ADV participation.
- **Case study:** Walk-forward + cross-sectional demean + cost adjustment is the minimal honest backtest; leakage is the silent killer.

Next practical

Practical 11: regulatory & compliance applications — structured extraction, auditability, and SR 11-7 model-risk obligations for LLM outputs.

APPENDIX

Stretch Problems

Extra / optional material — beyond the core lecture.

Appendix: Stretch Problem A — CVaR Budget

A position has daily returns approximated by a fat-tailed distribution with $\text{VaR}_{0.05} = -3.0\%$ and an expected shortfall (CVaR) of -4.5% in the worst 5%.

- A. Interpret both numbers in one sentence each.
- B. A textual risk signal v_{t-1} spikes; the GARCH update $\sigma_t^2 = \omega + \varphi_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 + \kappa v_{t-1}$ with $\kappa > 0$ raises σ_t by 20%. Qualitatively, what happens to $\text{CVaR}_{0.05}$ and to the CVaR-constrained weight on this position?
- C. Why is CVaR preferred to VaR for *optimization*? (Hint: coherence; convex/LP-representable.)

Appendix: Stretch Problem B — Minimum Track Record

Using the minimum track record length

$$T^* = 1 + \left(1 - \gamma_1 \text{SR}^* + \frac{\gamma_2 - 1}{4} (\text{SR}^*)^2\right) \frac{Z_{1-\alpha}^2}{(\text{SR}^*)^2},$$

with a *per-observation* (daily) $\text{SR}^* = 0.08$, skewness $\gamma_1 = -0.5$, kurtosis $\gamma_2 = 6$, and $Z_{0.95} = 1.645$:

- A.** Compute T^* in trading days, then in years ($\approx 252/\text{yr}$).
- B.** Repeat for a symmetric, normal benchmark ($\gamma_1 = 0, \gamma_2 = 3$). How much does the fat left tail inflate the required record?
- C.** What does a multi-year T^* imply for a strategy backed by only 18 months of OOS data?